
Research Paper: AI-Powered Movie Recommendation System via WhatsApp

Abstract

In today's content-saturated digital landscape, users often struggle to find movies that match their preferences. This paper presents a conversational, AI-powered movie recommendation system built using WhatsApp, Google Gemini, and Google Sheets. The system interprets natural language queries and delivers personalized suggestions instantly—without requiring app downloads or complex filters. Designed with no-code tools, it offers a scalable, beginner-friendly solution for student clubs and Gen Z users.

Introduction

Recommendation systems have become essential in guiding users through vast entertainment libraries. Traditional systems are embedded within apps and rely on rigid filters. This project explores a lightweight, conversational alternative using WhatsApp as the interface and Google Gemini for AI-driven understanding.

Objectives

- To build a movie recommendation system using WhatsApp and AI.
- To enable natural language queries for personalized suggestions.
- To use no-code tools for modular, scalable development.
- To create a replicable model for student clubs and educational showcases.

System Architecture

The system consists of five key components:

1. **WhatsApp Trigger** – Captures user messages.
2. **Google Gemini AI** – Extracts genre, mood, and keywords.
3. **Google Sheets** – Acts as a dynamic movie database.
4. **Memory Module** – Stores user preferences (optional).
5. **Send Message** – Delivers curated suggestions via WhatsApp.

All components are connected using Make.com or Pabbly, forming a clean, modular automation flow.

Working Methodology

- User sends a query like “Suggest a romantic movie with a twist.”
- Gemini interprets the message and extracts relevant parameters.

- It queries Google Sheets for matching entries.
- The system crafts a reply with the movie title, teaser, and cast.
- The reply is sent back via WhatsApp instantly.

Technologies Used

- **Google Gemini AI** – Natural language processing
- **Google Sheets** – Scalable database
- **Make.com / Pabbly** – No-code automation
- **WhatsApp Business API** – User interface
- **Prompt Engineering** – AI query optimization

Future Scope

- Actor/language-based filtering
- YouTube trailer integration
- Expansion to books and music
- Multilingual support
- Analytics dashboard for user insights

Conclusion

This project demonstrates how AI and no-code tools can create smart, scalable solutions for real-world problems. By combining conversational UX with modular design, it empowers students to build impactful tech with minimal resources.

References

1. Google Gemini – <https://deepmind.google/technologies/gemini>
2. WhatsApp Business API – <https://developers.facebook.com/docs/whatsapp>
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